AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 7 · STUDENT LESSON

Rows Columns and Data Types

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Before you start

Week 2: stages need agreed input formats.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Specify a table schema. Try the recall before opening notes.

Your learning target

I can specify a table **schema**.

I can check **type**s and **unit**s as well as names.

I can distinguish structural validity from truth.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Debugging Strategies. Give an example and a mistake that the idea helps you avoid.

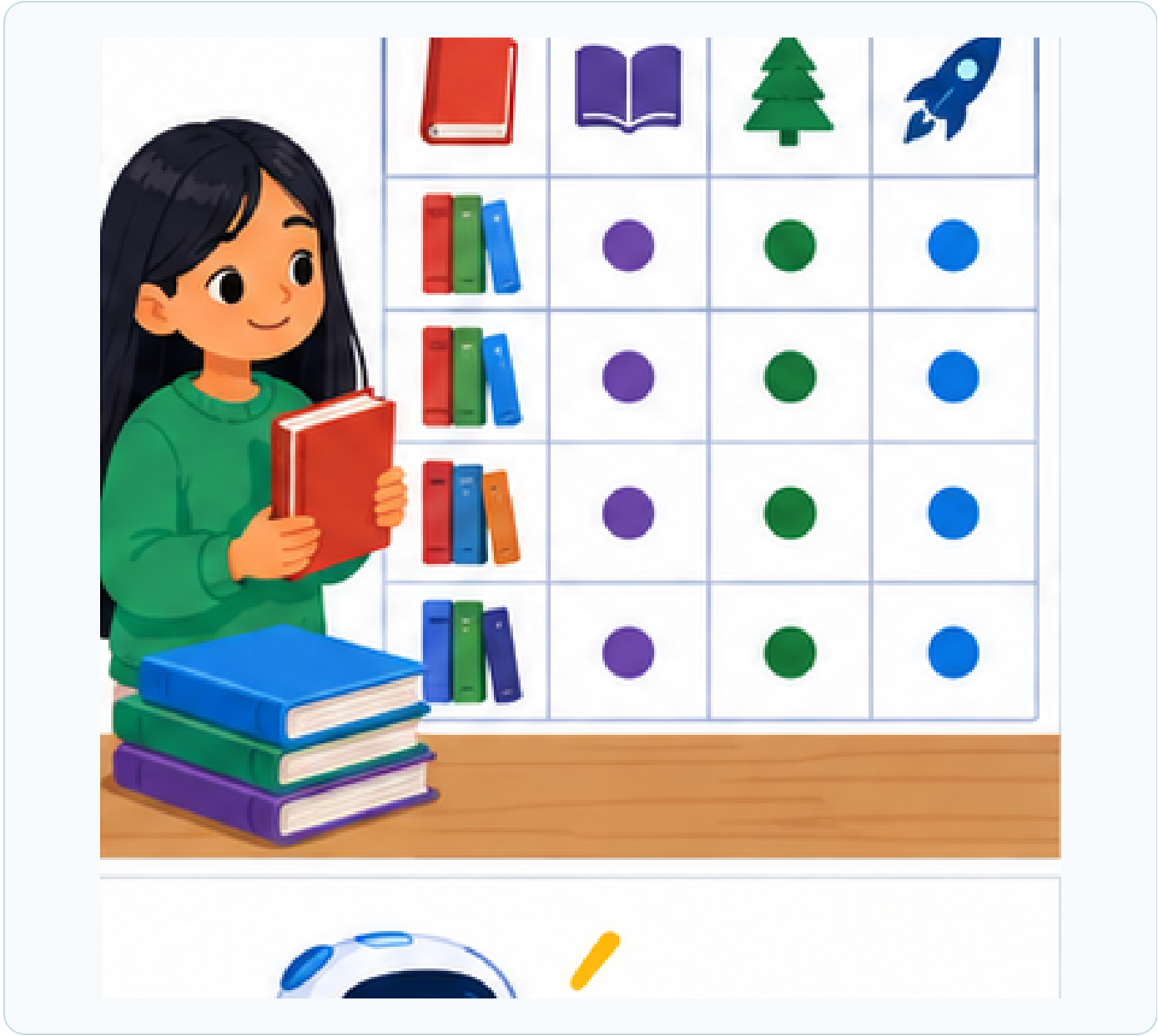
Without opening the old lesson, explain one key idea from Trace Tables. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Precision in Algorithms. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 7: Rows, **Column**s, and **Data Type**s

Structured tables preserve meaning through named fields and consistent **type**s.



THINKER MISSION

Repair a mixed-up class-pet dataset.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

What does row["used"] mean?

This week's end goal

By the end, I can read a table **schema** and distinguish a numeric value from its text representation.

STUDENT LESSON · 2 OF 18

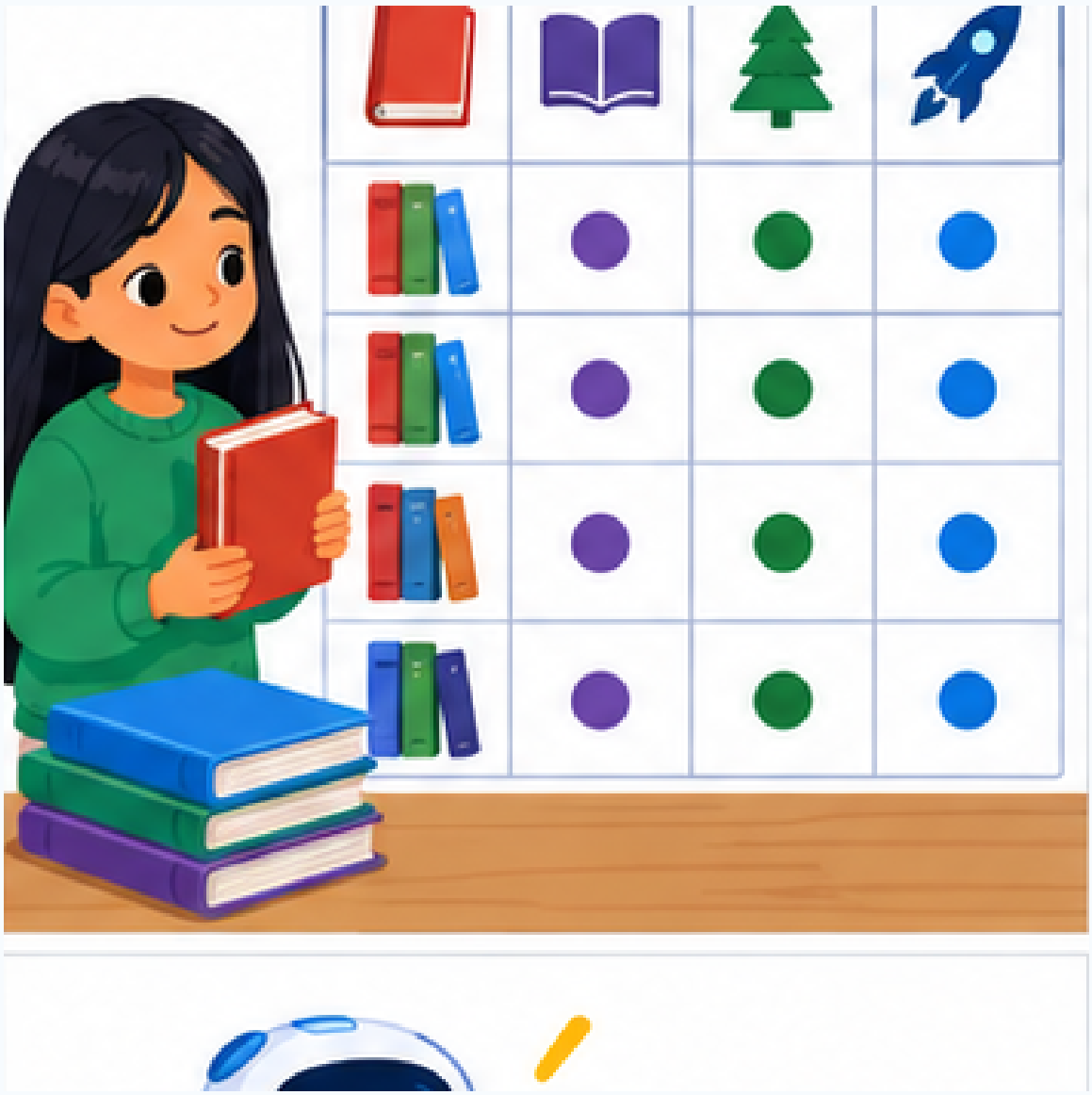
Notice the evidence

> Investigate the supplied Rows Columns and Data Types case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use rows, **column**s, and **data type**s to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

A fictional classroom logs the duration of each activity. Before comparing durations, define a **schema**: activity is nonempty text, minutes is an integer from 0 through 60, and each row describes one activity session.

These decisions make data interpretable.

STUDENT LESSON · 3 OF 18

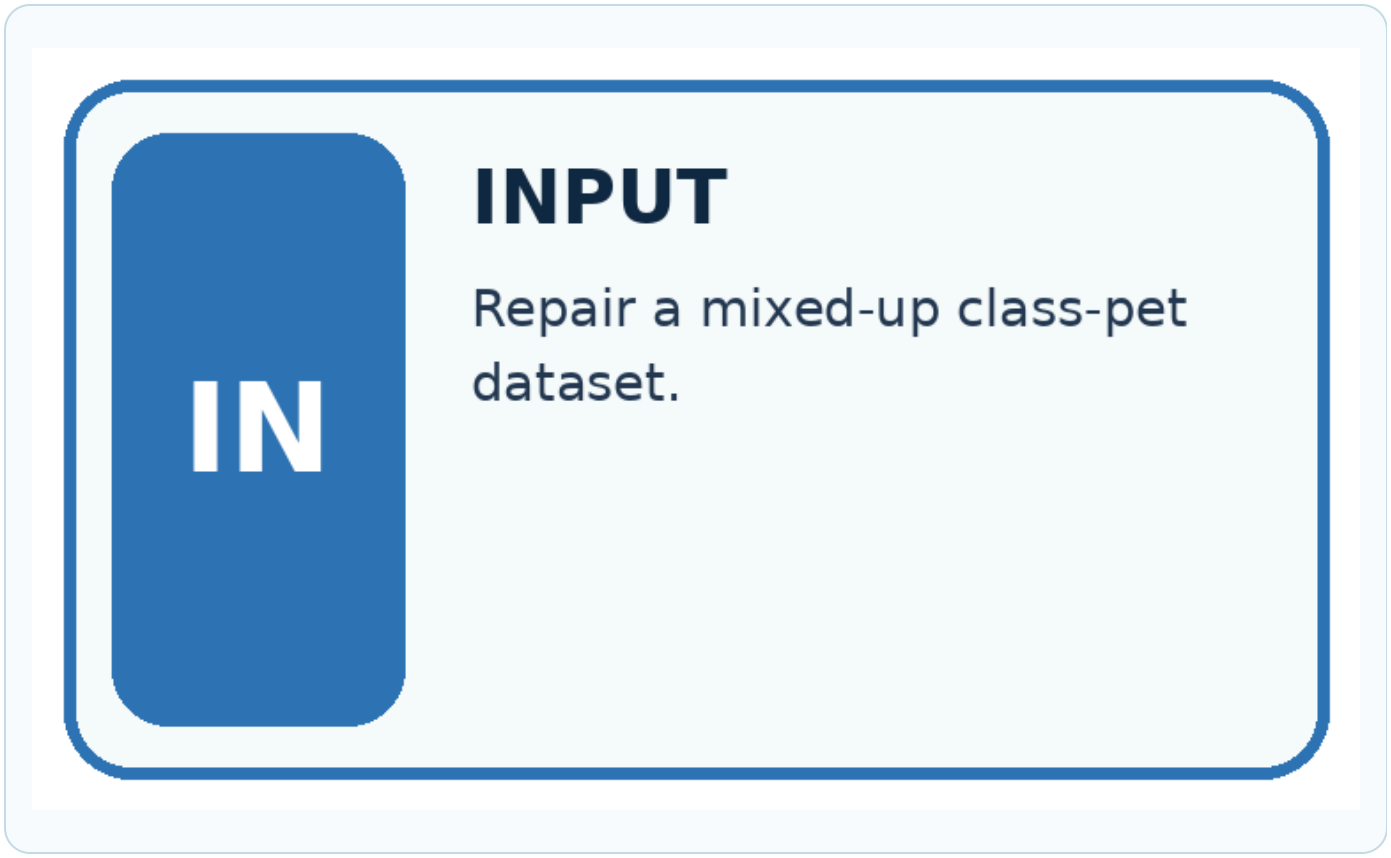
See how it works

> Follow the mechanism for Rows Columns and Data Types. Connect each labelled input, deciding step and result.

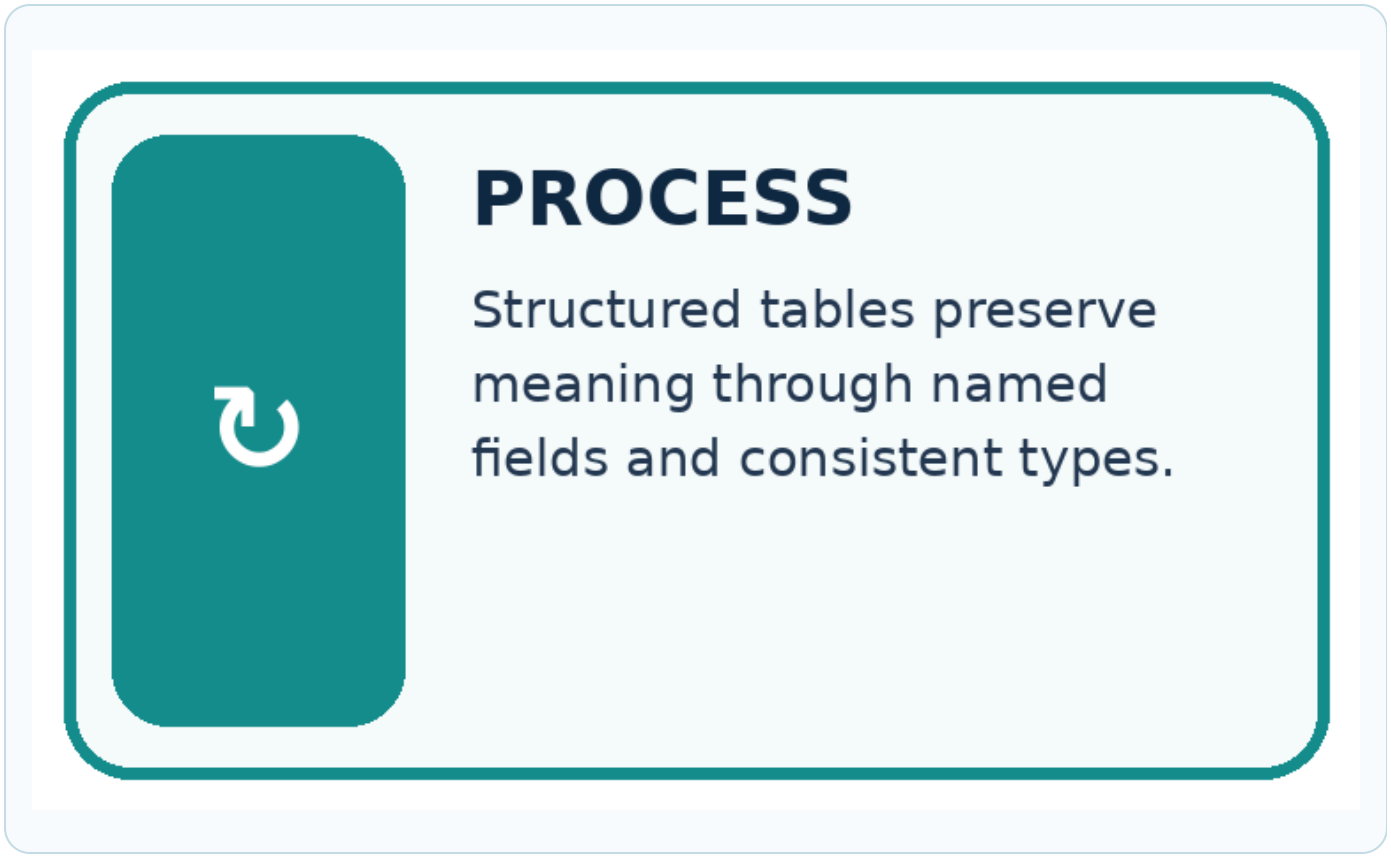
VISUAL MECHANISM

Input → **Process** → **Output**

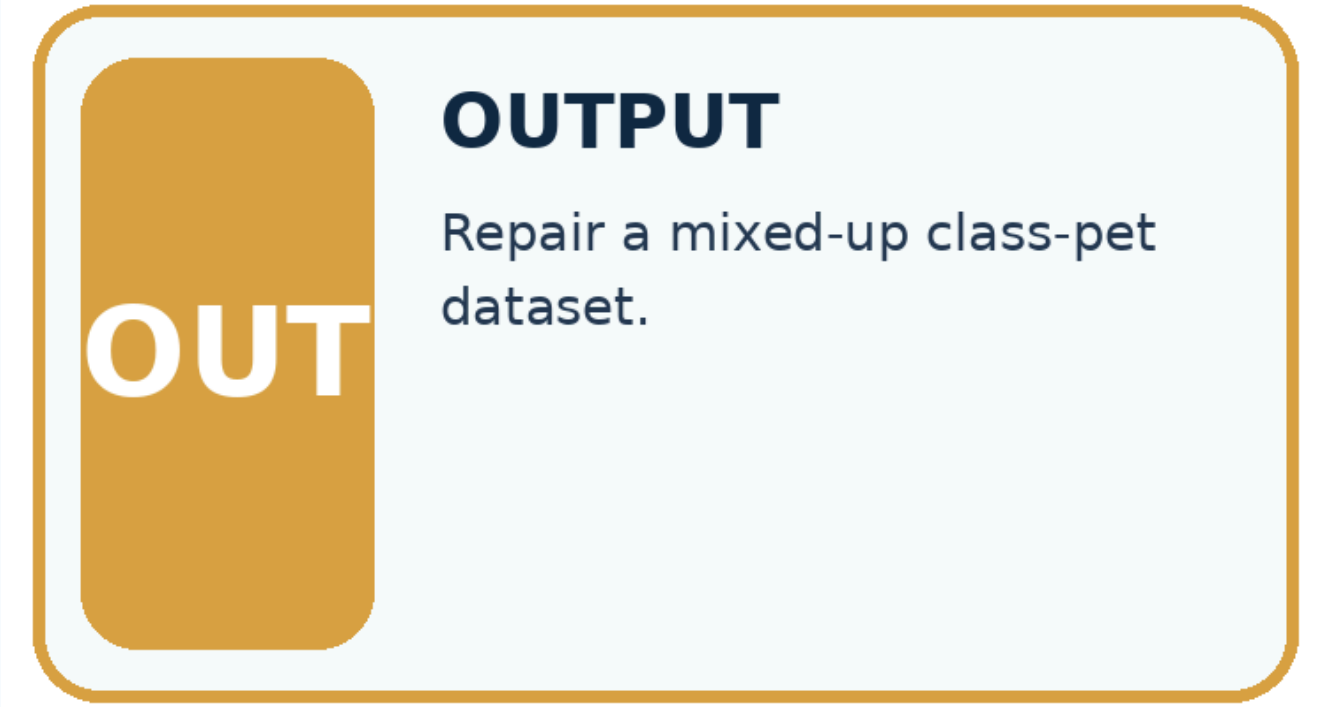
Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Repair a mixed-up class-pet dataset.

↓


PROCESS
Structured tables preserve meaning through named fields and consistent types.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

```
rows = [{"activity": "Reading", "minutes": 12},
        {"activity": "Sorting", "minutes": 0}]
for row in rows:
    valid = (type(row["activity"]) is str
             and len(row["activity"]) > 0
             and type(row["minutes"]) is int
```

```
            and 0 <= row["minutes"] <= 60)
print(valid)
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
activity	Nonempty text	Names the session activity
minutes	Integer 0 through 60	Duration in minutes

Read each row from left to right and connect the evidence to the stated rule.

The worked script prints True twice. **type**(value) is int checks the exact **type** here and avoids accepting a Boolean as a count. The chained comparison checks both boundaries.

The example assumes both keys exist; next week handles missing fields. A zero duration is allowed by this written **schema**.

Check a row in separate steps

A **schema** is a checklist for what a row must contain. First ask whether the activity name is text and is not empty. Next ask whether minutes is an integer.

Then ask whether that integer lies between 0 and 60, including both ends. Keep these checks separate when explaining a rejection.

The number 12 and the text "12" look similar on a page, but only the number meets this lesson's minutes rule.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

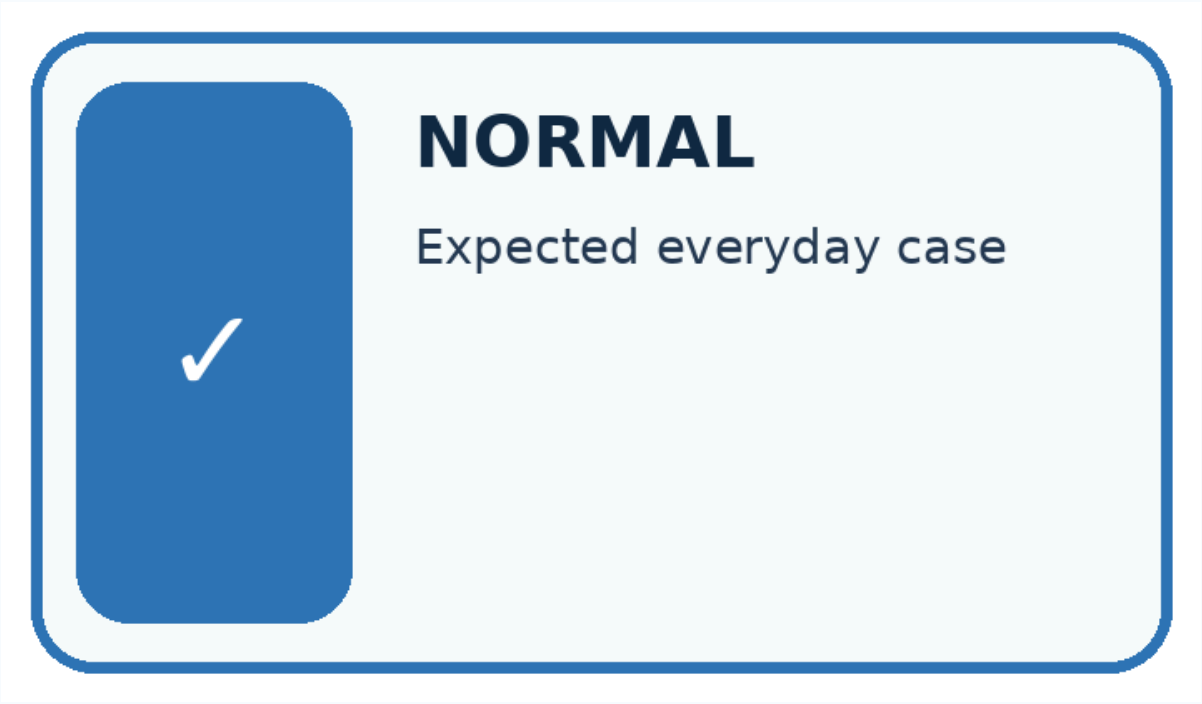
> Compare the supplied cases for Rows Columns and Data Types. Identify the detail that changes the conclusion.

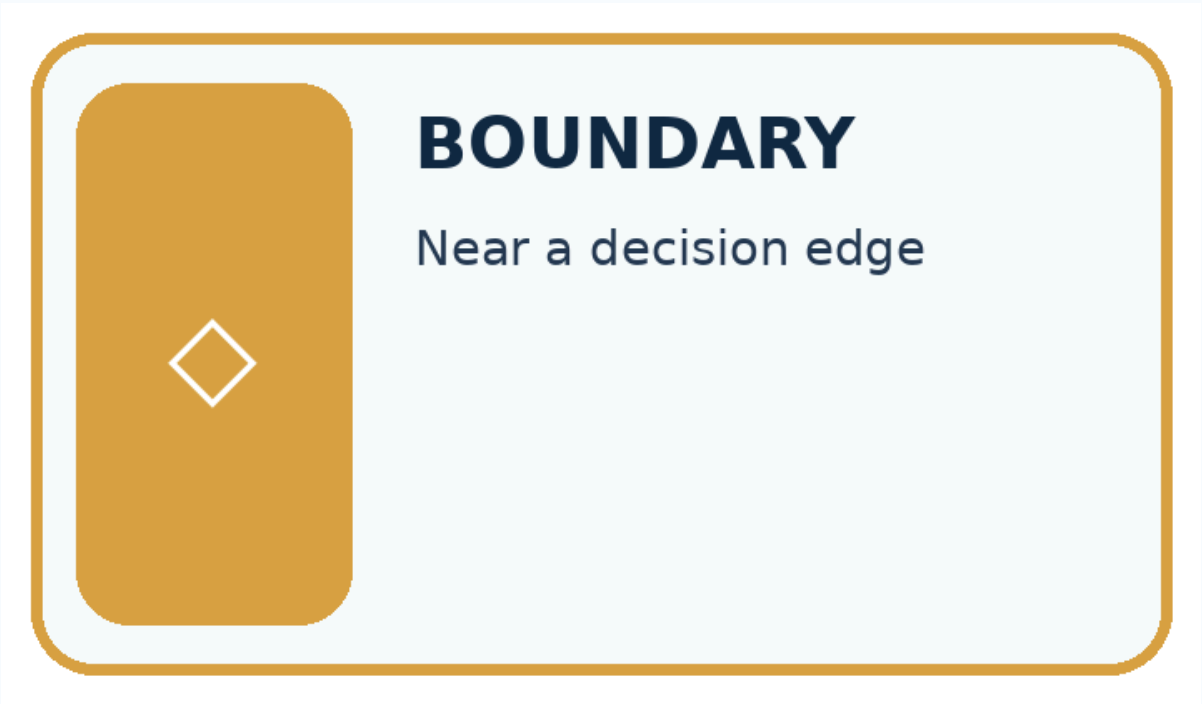
The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

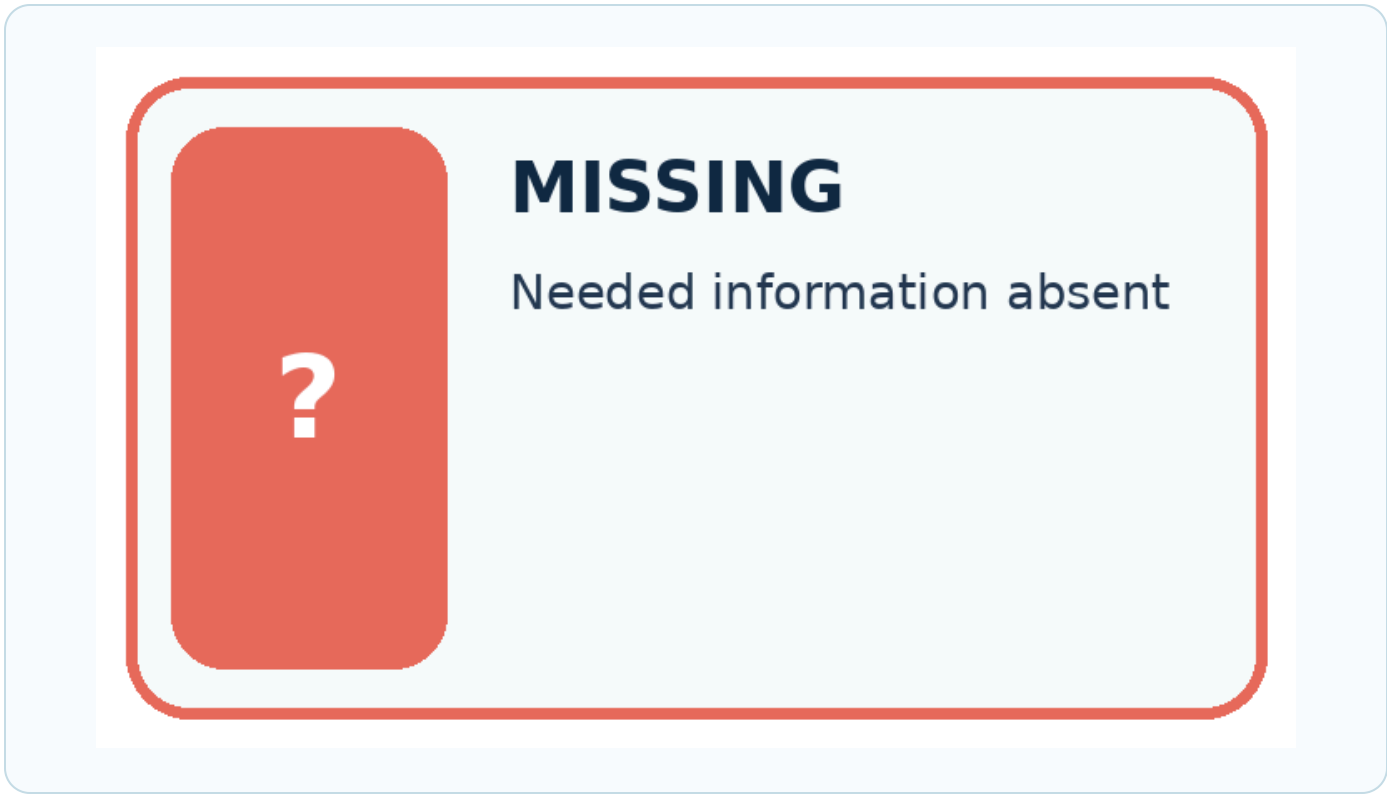
A system is not understood until it survives more than one comfortable example.





The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.


?
MISSING
Needed information absent


→
TRANSFER
Fresh situation or user

Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Change B's pages from "90" to integer 90 using an approved repair.

Predicted result

Prediction: B now satisfies the declared **schema**.

Why the result follows

Reason: Its id is nonempty and illustrated is Boolean; the page-**type** defect is removed.

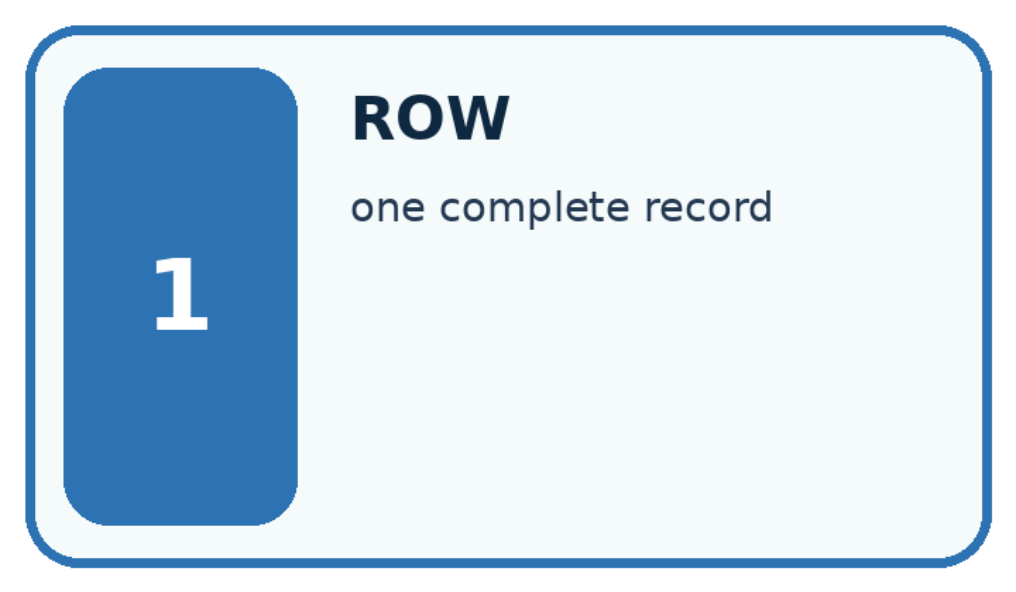
Words that unlock the idea

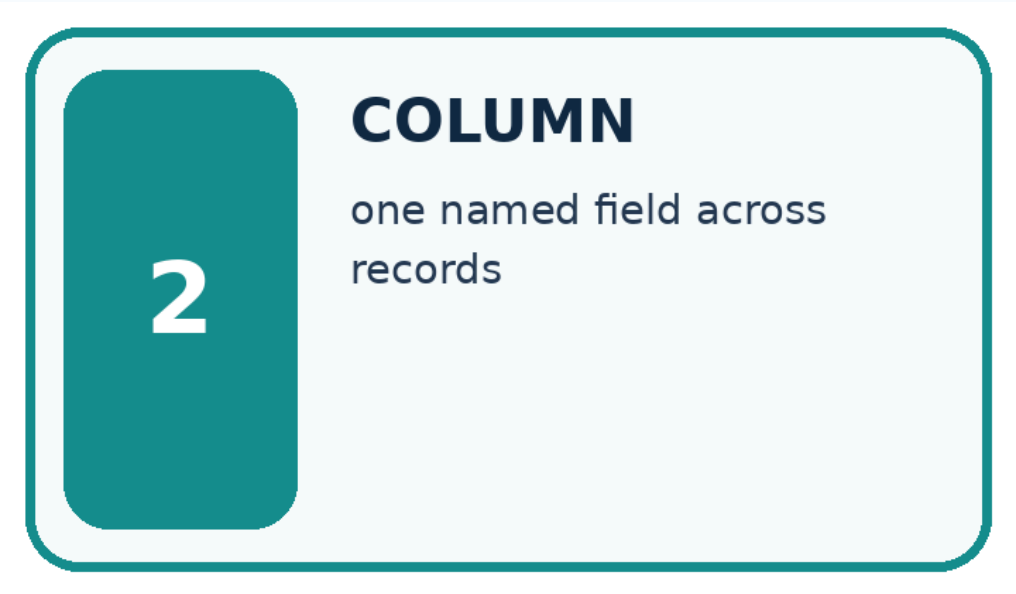
> Use the vocabulary of Rows Columns and Data Types to explain a concrete decision, rather than only memorising definitions.

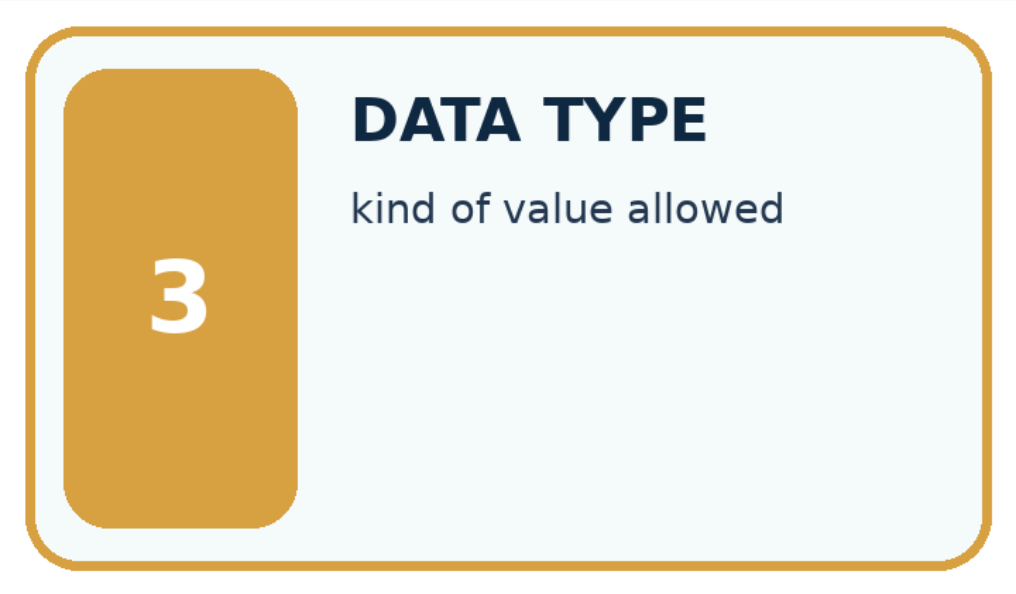
VOCABULARY LAB

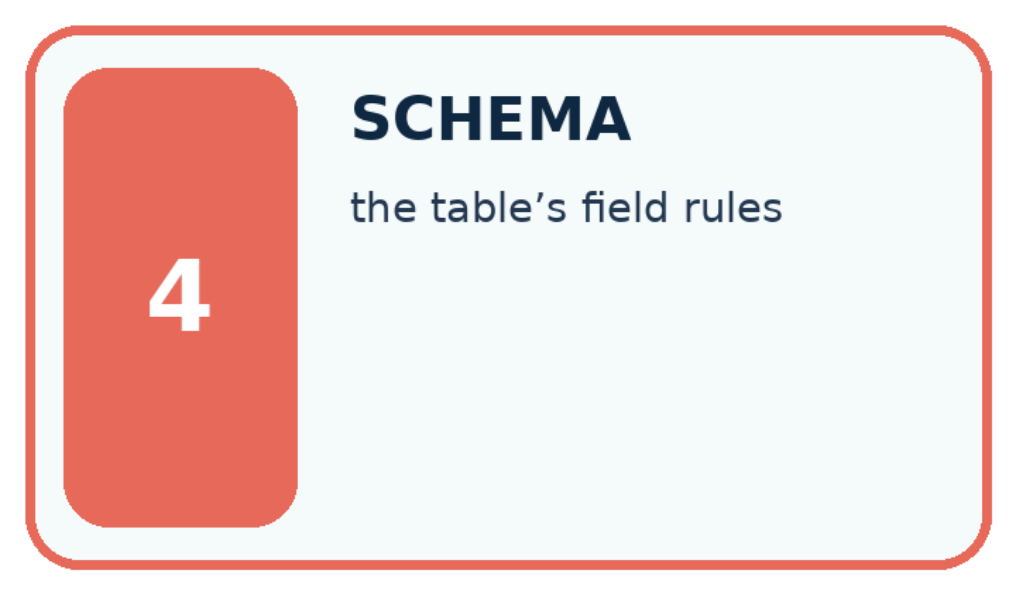
Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
COLUMN
one named field across records


3
DATA TYPE
kind of value allowed



Word	My own example	Not this / boundary
row	One record or example in a table.	
column	A table field containing the same kind of information across rows.	
data type	A category of values, such as number, text or Boolean, with defined operations.	
schema	A specification of required fields, types and structural rules for data.	

Words used in the closer explanation

Word	Meaning
Type	A category such as integer or text
Unit	The quantity scale attached to a number
Constraint	A rule a valid record must satisfy

STUDENT LESSON · 6 OF 18

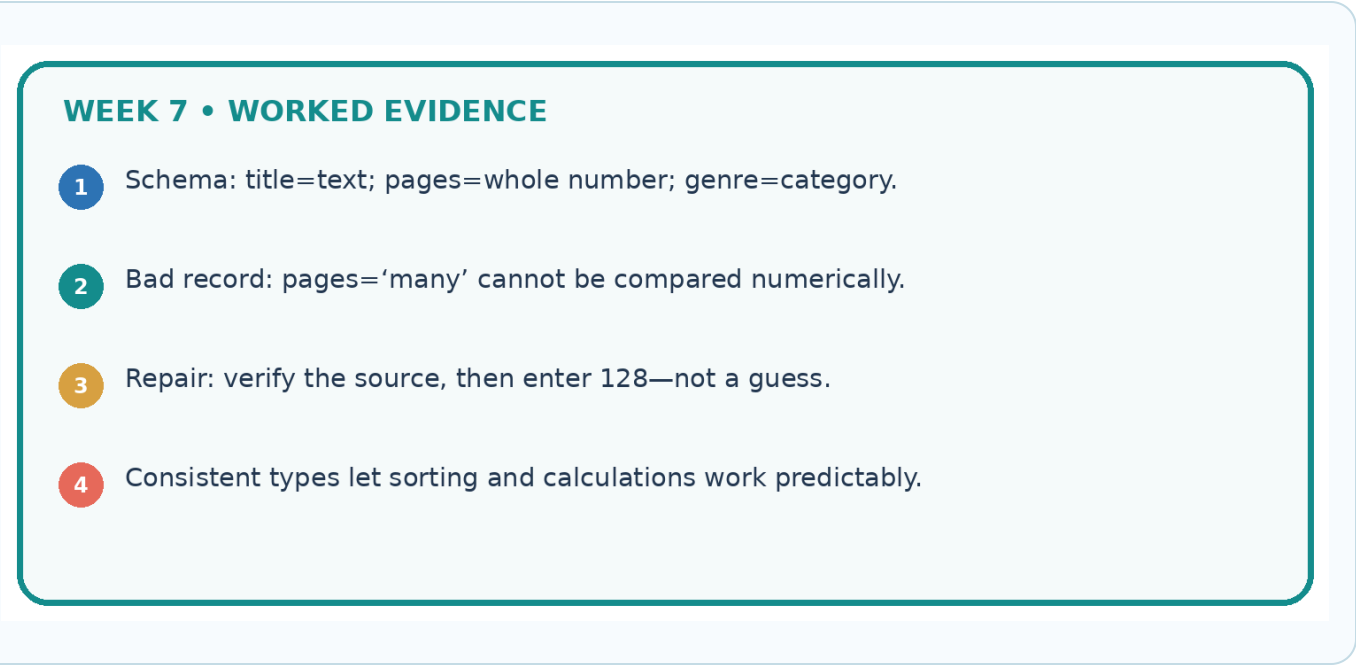
Follow a worked example

> Follow the worked trace for Rows Columns and Data Types, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. **Schema**: title=text; pages=whole number; genre=category.

2. Bad record: pages='many' cannot be compared numerically.

3. Repair: verify the source, then enter 128—not a guess.

4. Consistent **type**s let sorting and calculations work predictably.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Would minutes of 61, "12" and 12 each pass? Explain the type distinction.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Rows Columns and Data Types

A fictional approved book catalogue has three fields per row.

The facts we will use

Supplied facts

Schema: id is nonempty text; pages is an integer 1–300; illustrated is Boolean True or False.

Supplied facts

Rows A={id:"B7",pages:90,illustrated:True}; B={id:"B8",pages:"90",illustrated:False}; C={id:"",pages:80,illustrated:True}.

AI ACADEMY / AI-2/7

Rows Columns and Data Types

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Column contract

Field names identify values; order is not the only guide.



2. Validate A

B7 is text, 90 is allowed integer, True is Boolean: valid.



3. Validate B

Quoted "90" is text, so pages fails the integer contract.



4. Validate C

Empty id fails even though its other two values have allowed types.

CHANGE ONE THING

Change B's pages from "90" to integer 90 using an approved repair.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Column contract	Field names identify values; order is not the only guide.
2. Validate A	B7 is text, 90 is allowed integer, True is Boolean: valid.
3. Validate B	Quoted "90" is text, so pages fails the integer contract.
4. Validate C	Empty id fails even though its other two values have allowed types.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A **schema** makes row meanings and expected **type**s explicit. The string "90" looks like a number when printed but does not meet this strict integer **schema**. A repair can be authorised later, but must not be assumed silently.

Missing or empty identifiers also prevent tracing an item back to its source.

What this example does not establish

Keep the stated limit

Schema validity does not confirm that the book really has 90 pages or that its metadata is accurate.

Keep the stated limit

Read the labelled picture: Which two rows originally fail, and do they fail for the same reason?

Keep the stated limit

Change one thing: Change B's pages from "90" to integer 90 using an approved repair. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Rows Columns and Data Types claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: A spreadsheet cell's color tells the computer its data type. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

A row can satisfy every **schema** rule and still be a wrong **observation**. **Schema** validation checks our stated structure and **constraint**s, not whether someone measured honestly or selected a representative sample.

> **A row says Reading, 12 and passes. Does that establish the session really lasted 12 minutes?**

STUDENT LESSON · 8 OF 18

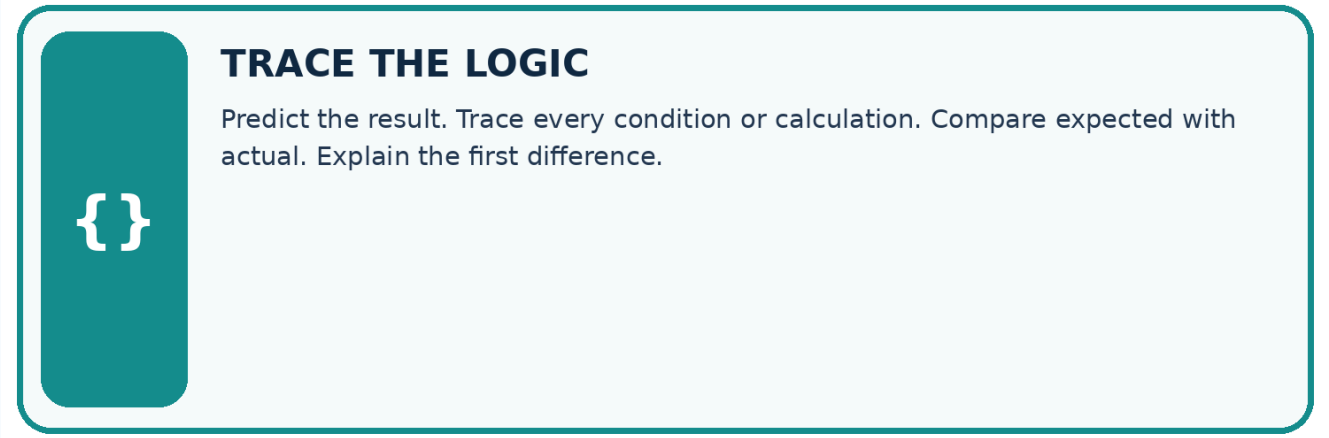
Practise together

> Practise Rows Columns and Data Types with support, then explain the deciding step in your own words.

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

> Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

If one row records 120 seconds in the minutes field, a simple numeric total becomes misleading. Correct the source or explicitly convert and document it; renaming a **column** alone does not convert values.

Practise with less help: Classify records

Use workbook W2 for the connected example “Rows **column**s and **schema**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Rows **column**s and **schema**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Which two rows originally fail, and do they fail for the same reason?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

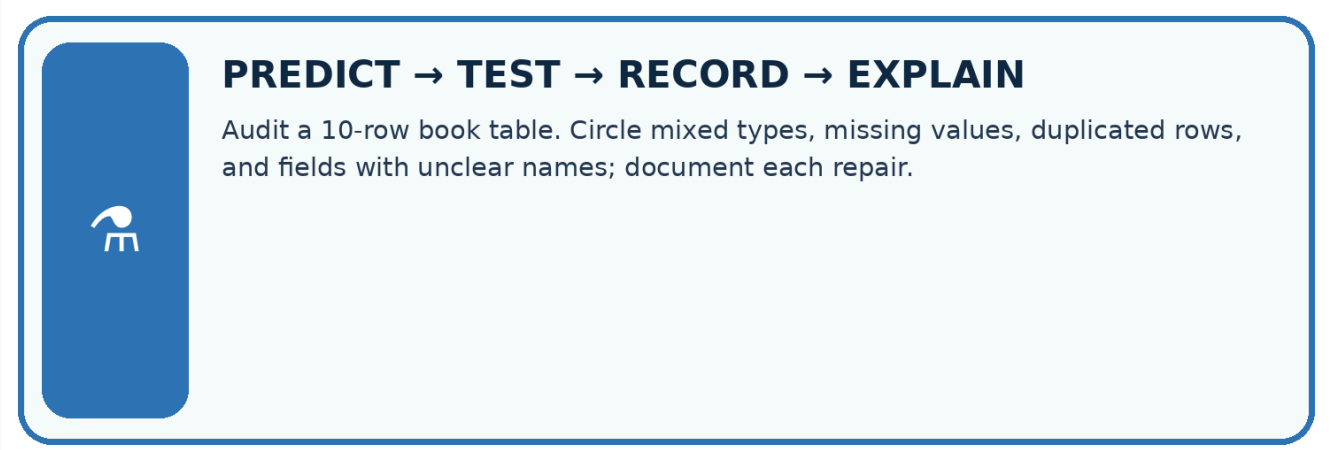
Run the investigation

> Investigate Rows Columns and Data Types. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Audit a 10-row book table. Circle mixed **type**s, missing values, duplicated rows, and fields with unclear names; document each repair.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

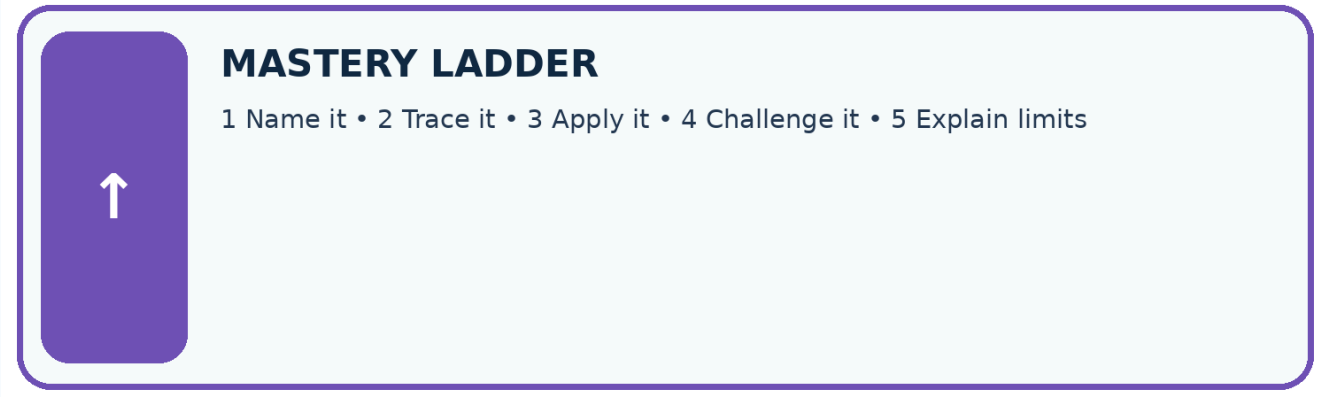
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Rows Columns and Data Types. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

> **Write or draw your reasoning and evidence.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Transparent Book Recommender. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Store one book per row and keep reader preferences in a separate record.

Trace the system

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Publish the recommender's row **schema** and validation results before ranking books, preserving repair provenance.

STUDENT LESSON · 12 OF 18

Show what you know

Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.

INDEPENDENT PROOF

Repair a mixed-up class-pet dataset.

Try a more demanding case

CHALLENGE Repair a mixed-up class-pet dataset.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Rows Columns and Data Types case for supported practice. Repeated examples do not become fresh assessments.

Week 7 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Fields are Title, Pages and Borrowed; a row contains River, ninety and yes.

The question

Assign intended **type**s and a repair.

Worked explanation

Text, numeric and Boolean. Confirm ninety means 90, then convert and record the change; use a consistent Boolean representation.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

A field has a meaning and a type

Evidence or stage	Value or result
Title	Text → River
Pages	Number → 90 after checking source
Borrowed	Boolean → true or false

Fields are Title, Pages and Borrowed; a row contains River, ninety and yes.

Assign intended **type**s and a repair.

Text, numeric and Boolean. Confirm ninety means 90, then convert and record the change; use a consistent Boolean representation.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Design a schema independently

For a fictional shelf audit, define label and book_count fields. Require a nonempty text label and an integer count from 0 to 40. Give two valid and two invalid records with reasons.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 7 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A table row is book_id B7, pages 84, available true. Name the **type** of each field and explain why pages stored as eighty-four complicates calculation.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Store one book per row and keep reader preferences in a separate record.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

A row contains title='Sky', pages='120', available='false'. Explain why treating nonempty text as true can misread availability. Propose explicit conversions and a rejected-**input** case.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Rows Columns and Data Types, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional approved book catalogue has three fields per row.

By the end, I can	How I will show it
Specify a table schema	Explain the deciding step, show a trace or test, and state one limit.
Check types and units as well as names	Explain the deciding step, show a trace or test, and state one limit.
Distinguish structural validity from truth	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Rows Columns and Data Types without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Specify a table **schema**

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional approved book catalogue has three fields per row.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Empty id fails even though its other two values have allowed **type**s.

1. **Column** contract

2. Validate C

3. Validate B

4. Validate A

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Validate C Empty id fails even though its other two values have allowed **type**s. A **schema** makes row meanings and expected **type**s explicit.

The string "90" looks like a number when printed but does not meet this strict integer **schema**. A repair can be authorised later, but must not be assumed silently.

Missing or empty identifiers also prevent tracing an item back to its source.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change B's pages from "90" to integer 90 using an approved repair. Which conclusion is supported by the supplied facts?

1. B now satisfies the declared **schema**.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: B now satisfies the declared **schema**. Its id is nonempty and illustrated is Boolean; the page-**type** defect is removed.

Repair a missing building block

Review Debugging Strategies (ai-2/6). Goal: Design revealing tests for a function. Short practice: Which test most directly isolates the handling of loose books, and what should it produce?.

Return to this week and explain what you changed.

Review Trace Tables (ai-2/4). Goal: Trace variable values before and after each loop iteration. Short practice: Move print inside the original loop. What sequence is displayed?.

Return to this week and explain what you changed.

Review Precision in Algorithms (ai-2/1). Goal: Translate a short paper calculation into a Python script. Short practice: If print(total) moves before total=total+2, what gets printed for the original **input**?.

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

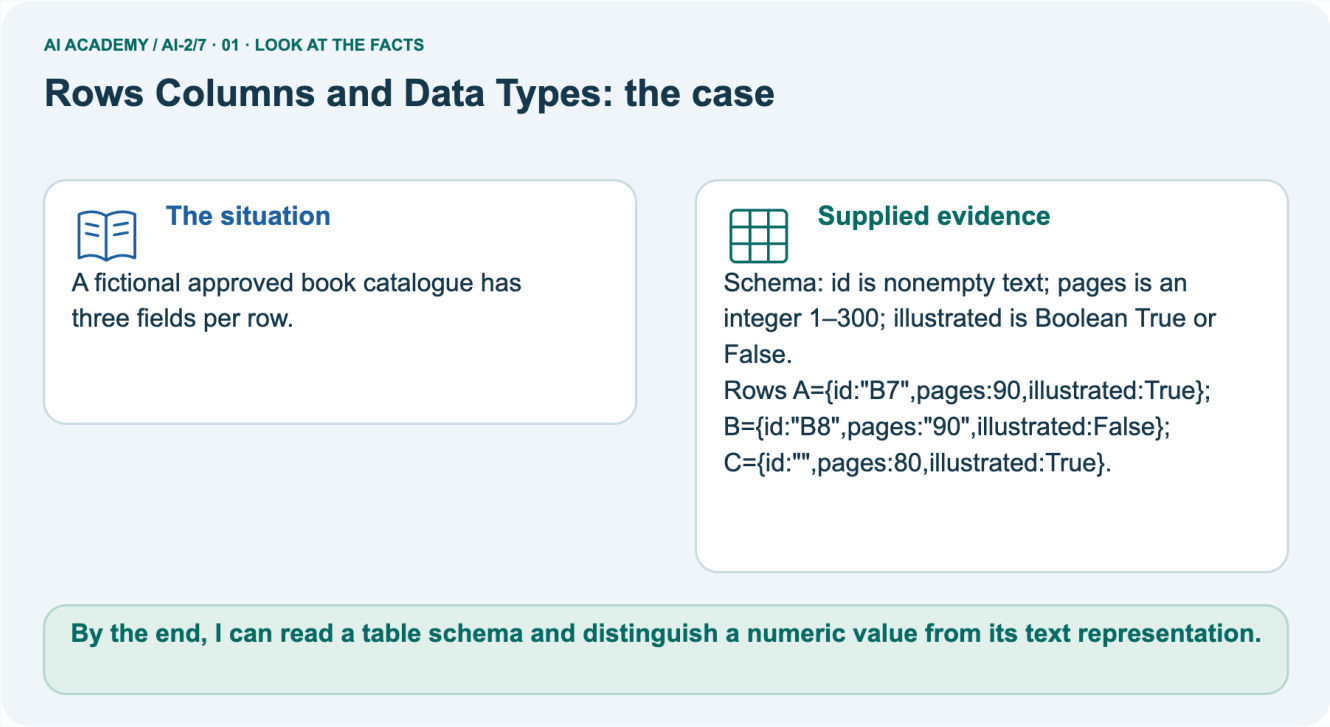
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Rows Columns and Data Types graphic using its labels, deciding evidence and relevant limitation.

01. Rows Columns and Data Types: the case



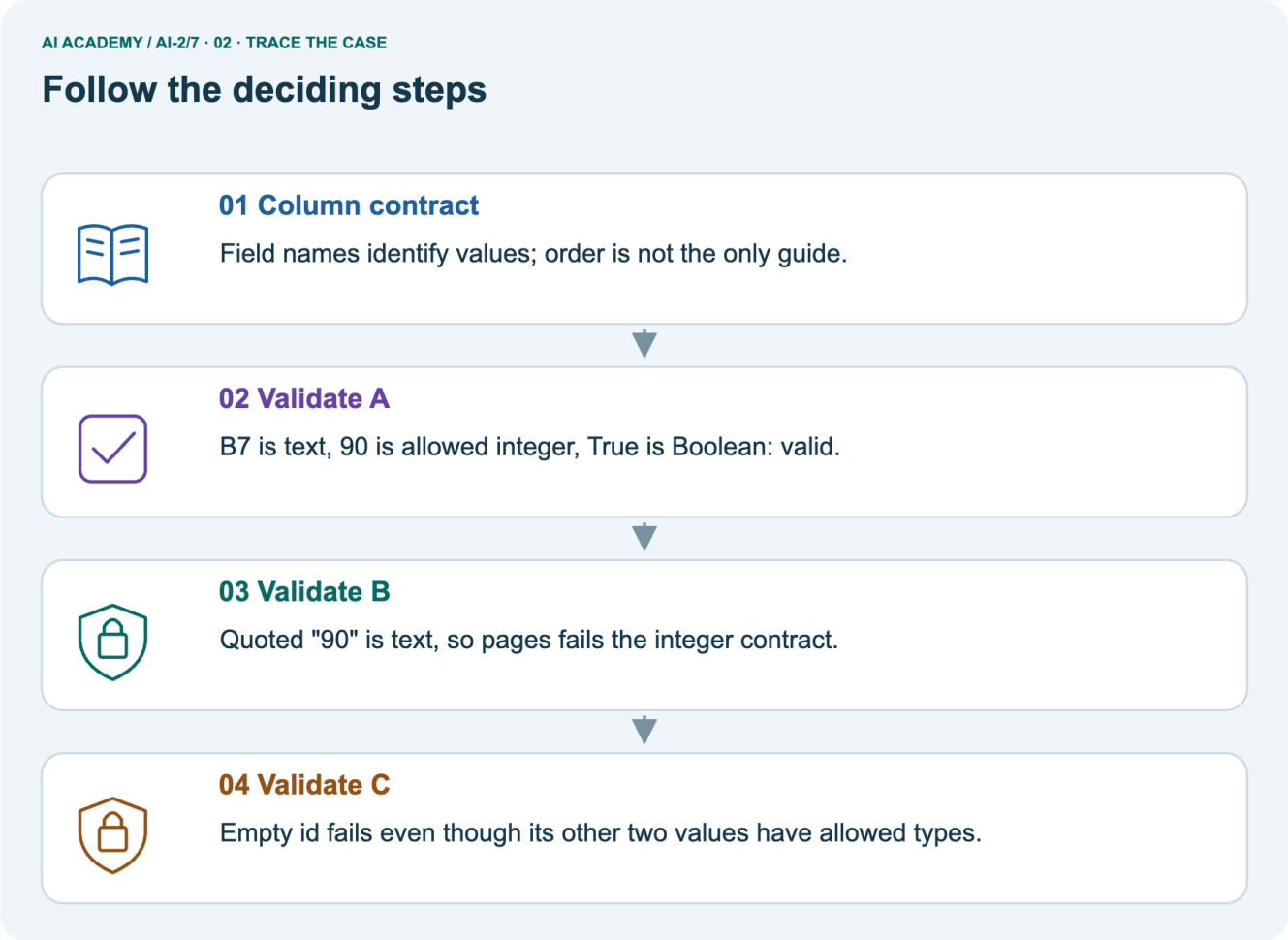
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional approved book catalogue has three fields per row. Supplied facts: **Schema**: id is nonempty text; pages is an integer 1–300; illustrated is Boolean True or False.

Rows A={id:"B7",pages:90,illustrated:True}; B={id:"B8",pages:"90",illustrated:False}; C={id:"",pages:80,illustrated:True}.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

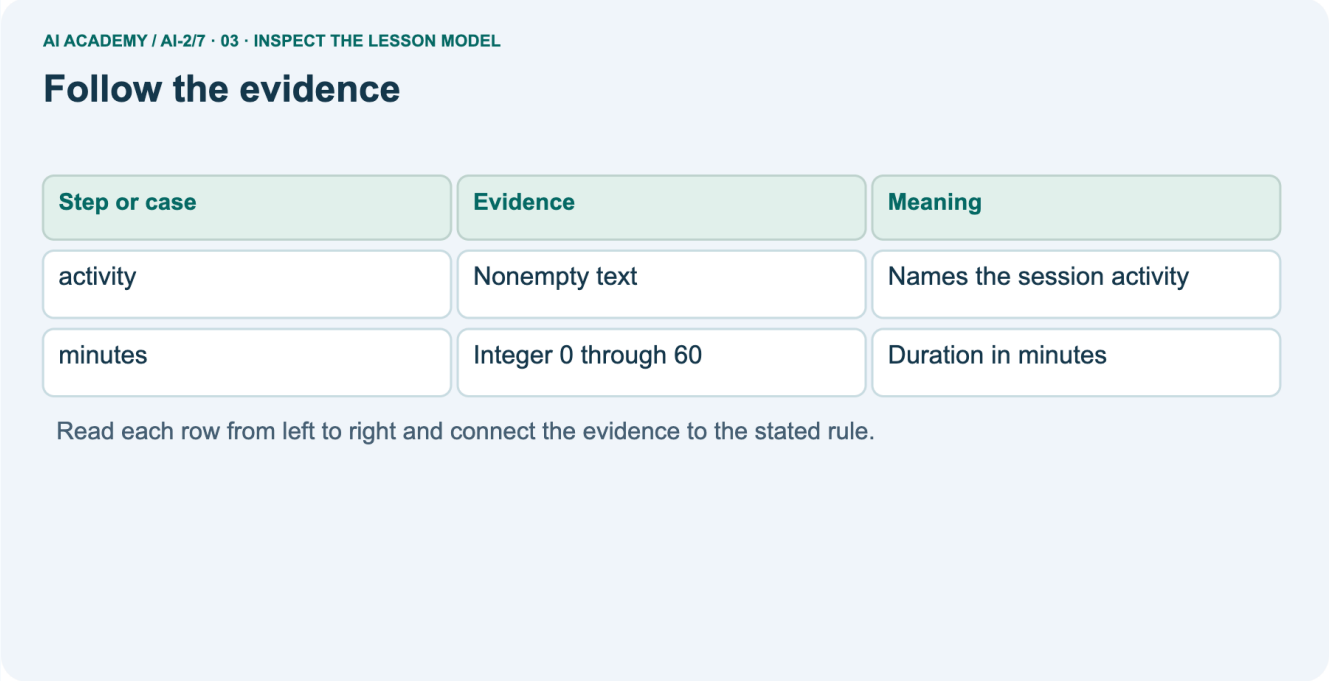


Follow the labelled picture with the matching explanation. Open the vector diagram

Column contract: Field names identify values; order is not the only guide. → Validate A: B7 is text, 90 is allowed integer, True is Boolean: valid. → Validate B: Quoted "90" is text, so pages fails the integer contract. → Validate C: Empty id fails even though its other two values have allowed **type**s.

Point to each numbered stage and explain which recorded **input** it uses.

03. Follow the evidence

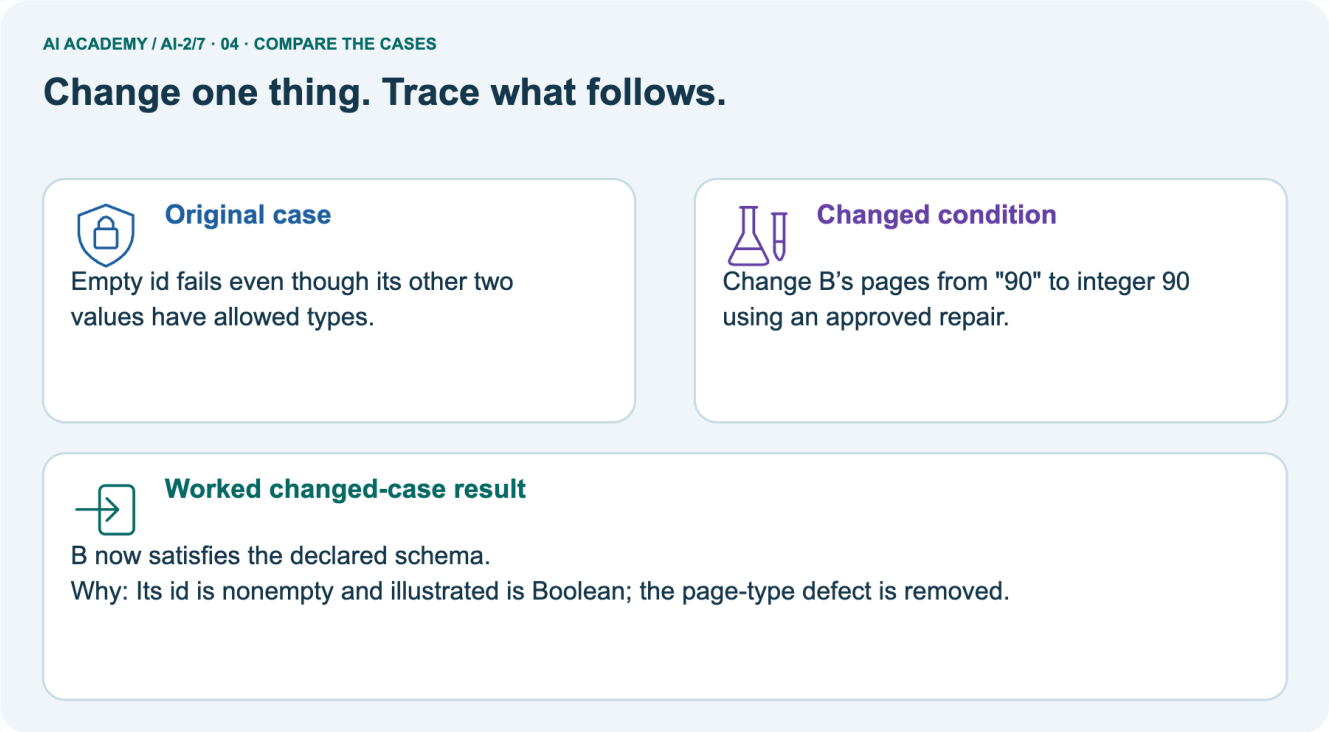


Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. activity | Nonempty text | Names the session activity. minutes | Integer 0 through 60 | Duration in minutes

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



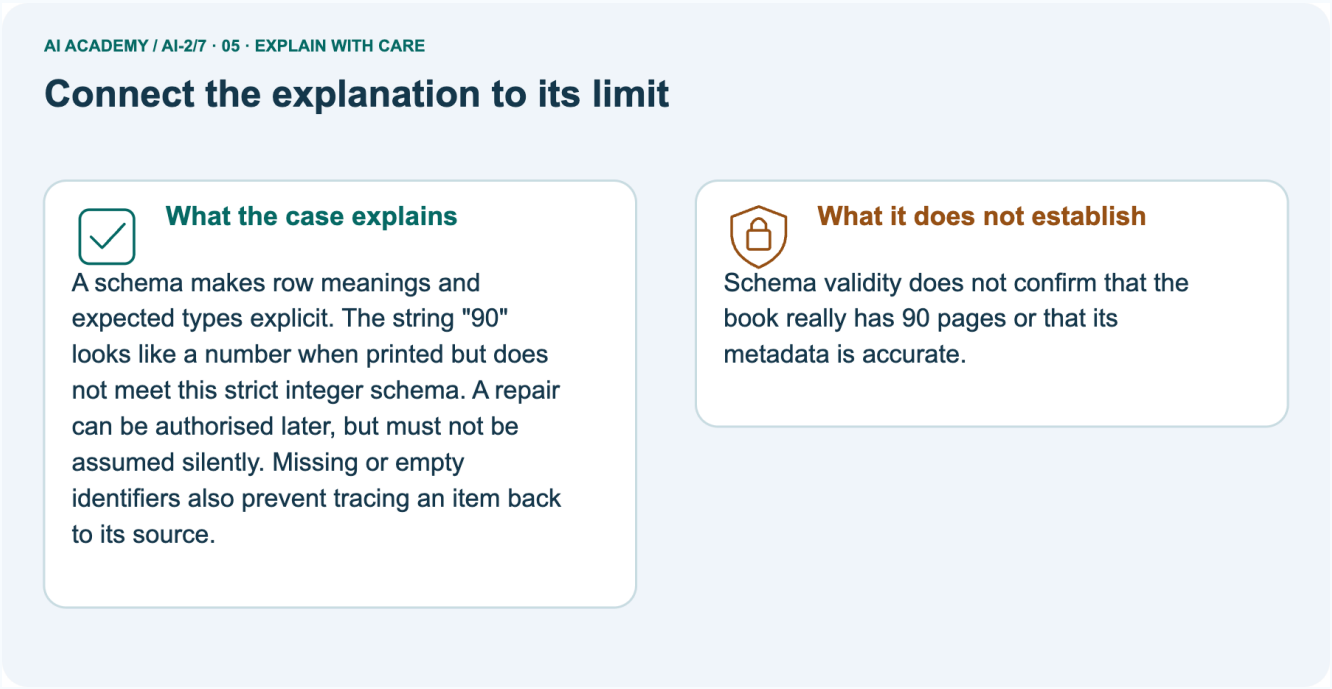
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Empty id fails even though its other two values have allowed **type**s. Changed condition: Change B's pages from "90" to integer 90 using an approved repair. Expected result: B now satisfies the declared **schema**.

Reason: Its id is nonempty and illustrated is Boolean; the page-**type** defect is removed.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



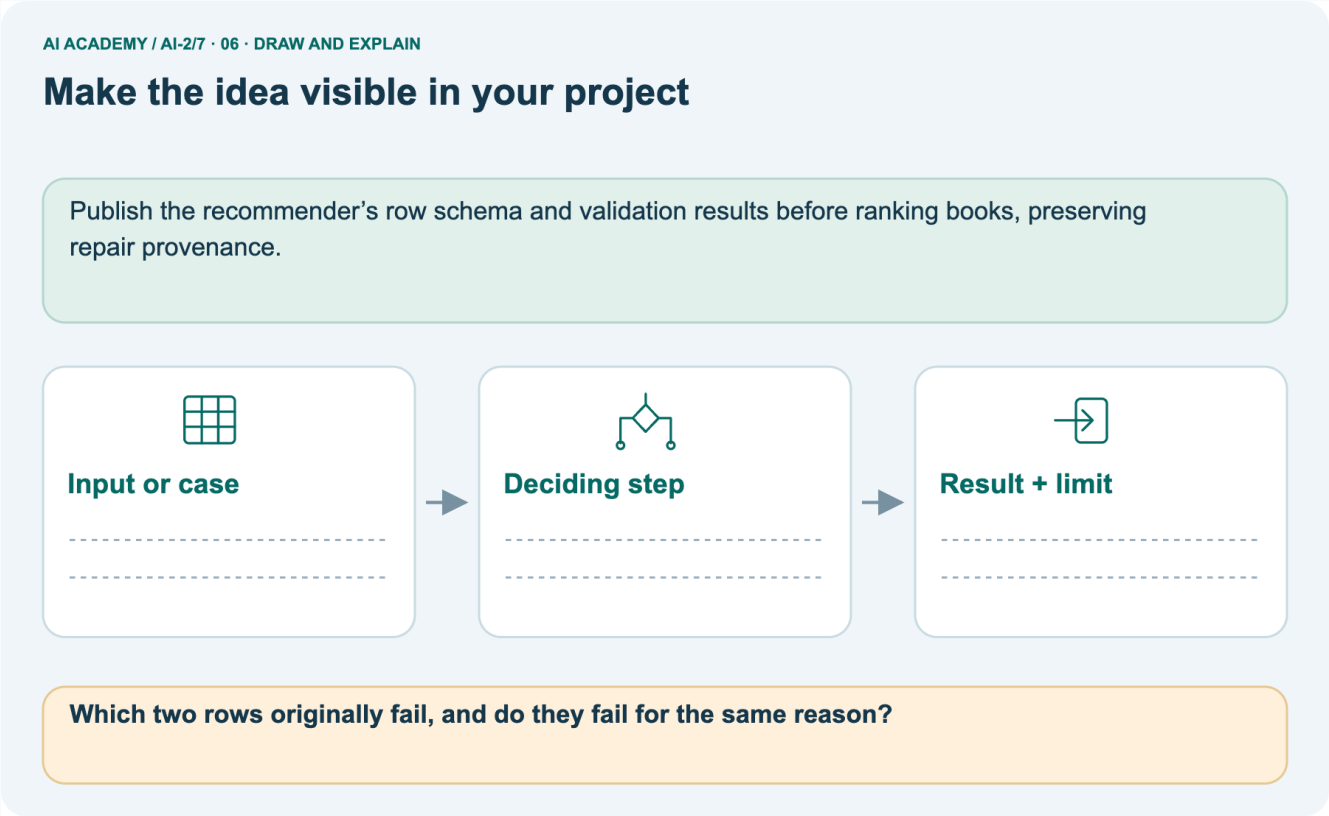
Follow the labelled picture with the matching explanation. Open the vector diagram

A **schema** makes row meanings and expected **type**s explicit. The string "90" looks like a number when printed but does not meet this strict integer **schema**. A repair can be authorised later, but must not be assumed silently.

Missing or empty identifiers also prevent tracing an item back to its source. Limit: **Schema** validity does not confirm that the book really has 90 pages or that its metadata is accurate.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Publish the recommender's row **schema** and validation results before ranking books, preserving repair provenance. Draw your **input**, deciding step, result and limit.

Supported question: Which two rows originally fail, and do they fail for the same reason?

Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.